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Fall 2024 MAT 2362
Instructor: Kirill Zaynullin

Midterm Test 2 (v1)

First name: _____ Last name: _____

Student number:

Please, read the following instructions carefully:

- This is a **closed book exam**. All electronic communication devices (e.g. cell phones, calculators, laptops, tablets) must be turned off and **put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device (even if it is switched off), academic fraud allegations will be filed.
- Fill in your first name, the last name and the student number. **Do not detach** the pages of this examination.

This space is reserved for marker

Nr	*1*	*2*	*3*	*4*	*5*	*6*	Total
Mark							
Out of	4	3	3	2	4	4	20

1. Find the supremum and infimum (if exist) of the following subsets of $\mathbb{R}$ (4pt)

$$\{\tfrac{1}{6n} + (-1)^n : n \in \mathbb{Z},\, n \geq 1\}.$$

(If the supremum or infimum does not exist in this case, explain why. If it exists, provide detailed arguments.

A correct answer without any explanation is 0pts.)

Solution:

2. Let $(X, \preceq)$ be a totally ordered set. Prove that if x_o is the minimal element, then it is also a minimum element. (3pt)

Proof:

3. Prove that the set $S = \{x\colon \sin(19x) \in \mathbb{Q}\}$ is countable. (3pt)

Answer:

4. Prove that $6\mathbb{Z} \sim 19\mathbb{Z}$ by means of giving explicit bijections. (2pt)

Proof:

5. Prove that $\text{Func}(\mathbb{N}, \mathbb{N}) \sim \mathbb{R}$ (Hint: use the graph of the function f to construct an injection). `(4pt)`

Proof:

6. Let A be an infinite set and let $\mathfrak{p}_{\text{fin}}(A)$ be the set consisting of all finite subsets of A that is

$$\mathfrak{p}_{\text{fin}}(A) = \{X \subseteq A \colon X \text{ is finite}\}.$$

Prove that $\mathfrak{p}_{\text{fin}}(A) \sim A$. (4pt)

Proof:

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